UNIVERSITY OF PETRA

Faculty of Information Technology - Department of Data Science and Analytics

Apex Performance Analytics: Automotive Tuning & ROI Optimization

A Unified Business Intelligence and Machine Learning Framework for High-Performance Vehicle Modification

A Graduation Project Report submitted in partial fulfillment of the requirements for the Degree of Bachelor in Data Analytics / Business Intelligence

Prepared by:	Abdullah Fleifel • [Student Name 2] • [Student Name 3]
Supervised by:	[Supervisor Name]
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Project Repositories & Deliverables:

GitHub Repository: https://github.com/uopetra/GP_BI20252.git
Google Drive Link: [Insert Google Drive Link Here]

Chapter 1: Abstract & Executive Summary

1.1 Project Overview

The aftermarket automotive performance tuning industry has historically operated as an unscientific environment dominated by empirical trial-and-error methodologies and mechanic intuition. The Apex Performance Analytics project addresses these multi-disciplinary challenges by establishing a rigorous, unified framework that bridges the gap between mechanical engineering boundaries, data science, and Business Intelligence (BI). The core mission is to direct mechanical engine modifications securely and accurately calculate the Return on Investment (ROI) based on thermodynamic constraints.

1.2 Problem Statement

The core problems this project solves are twofold. First, the significant financial waste resulting from inefficient capital allocation ("overbuilding") where expensive aftermarket parts are purchased without yielding proportional horsepower gains. Second, the extreme mechanical risk ("underbuilding") which leads to catastrophic, irreversible engine failure when forced induction boost pressure is increased without considering fuel octane quality and the tensile strength of internal engine components. The fragmentation of tuning data across unverified forums prevents a unified reference for engineering decisions.

1.3 Project Objectives

- To synthesize and process a high-fidelity dataset of 25,000 unique tuning configurations mapped across 94 distinct mechanical and financial features.
- To deploy a robust ETL data engineering pipeline using Python and Pandas, ensuring automated data cleaning, missing value imputation, and feature scaling.
- To train and evaluate an advanced Machine Learning ensemble capable of classifying engine safety status into three tiers (Safe, Risky, Critical Failure) based on deterministic physics.
- To design an Interactive Dashboard in Power BI utilizing DAX calculations to simulate financial ROI and identify optimal tuning thresholds.

Chapter 2: Data Engineering & Feature Processing

2.1 Dataset Composition

The dataset utilized in this project, titled `The_Flawless_Tuning_ERP_V20_Fixed.csv`, is highly comprehensive. It contains exactly **25,000 records** representing independent vehicle tuning builds. Each record is mapped across **94 distinct columns (features)** that cover engine platforms, forced induction specifications, target boost pressure (Target_Boost_Bar), fuel type, cooling efficiency, and total build costs (ranging comprehensively from $2,541 to $50,433).

2.2 Feature Engineering: Safety Threshold Logic

A primary challenge in the raw data was the absence of negative failure classifications, which would prevent a supervised learning algorithm from identifying mechanical risks. To rectify this, a custom Python algorithm was deployed to synthesize the `Safety_Status` target variable based on strict, deterministic mechanical engineering rules. Specifically, the logic cross-referenced Target Boost (Bar) against Fuel Type (e.g., 98 Octane vs. Methanol) and Engine Internals (OEM vs. Forged/CP-Carrillo). The resulting, highly realistic safety distribution yielded:

- **Safe:** 21,521 vehicles
- **Risky:** 1,888 vehicles
- **Critical Failure:** 1,591 vehicles

2.3 Data Cleaning and Normalization

Data preprocessing was orchestrated using Python's `pandas` and `scikit-learn` libraries. Missing numerical values (Nulls) were treated using programmatic Mean Imputation to preserve the shape of the data distribution, while missing categorical strings were replaced with standard defaults. To transition text columns into mathematical vectors, Label Encoding was applied. Furthermore, to stabilize algorithm weights for features evaluated on vastly different mathematical scales (e.g., $50,000 cost vs. 2.0 Bar Boost), **Min-Max Normalization** was executed, scaling all continuous inputs into a strict [0, 1] interval.

Chapter 3: Machine Learning Architecture

3.1 Algorithm Selection

A **Random Forest Classifier** was selected as the optimal algorithm to predict the multi-class target variable. This ensemble learning method constructs a multitude of independent decision trees and merges them together to get a more accurate and stable prediction. It is exceptionally capable of mapping complex, non-linear physical interactions—such as the volatile thermodynamic relationship between boost pressure and fuel octane ratings—while inherently mitigating the risk of overfitting on a dataset with 94 dimensions.

3.2 Dataset Partitioning

To guarantee valid academic evaluation and eliminate data leakage, the dataset was partitioned using an 80/20 Stratified Split. The Random Forest model utilized **80% (20,000 records)** for the training phase. The remaining **20% (5,000 records)** were isolated as a blind holdout testing set to evaluate real-world prediction accuracy.

Figure 3.1: Python Data Pipeline Flowchart

```
[Raw Data: 94 Columns] -> [Engineering Rules: Safety_Status Gen] ->
                      [Impute Nulls (Pandas)]
-> [Label Encoding] -> [Min-Max Scaling] -> [Stratified Split (80/20)]
-> [Random Forest Ensemble] -> [Holdout Evaluation] -> [Power BI Export]
```

Figure 3.1: A flowchart tracing the sequential ETL and modeling operations.

Chapter 4: Implementation Code

4.1 Model Execution Script

The predictive model was executed via the following Python script, validating the data engineering and algorithmic approach:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix,
accuracy_score

# Load the fixed dataset (25,000 rows, 94 columns)
df = pd.read_csv('The_Flawless_Tuning_ERP_V20_Fixed.csv')

# Separate Features (X) and Target (y)
X = df.drop('Safety_Status', axis=1)
y = df['Safety_Status']

# Apply Label Encoding to categorical data
le = LabelEncoder()
for col in X.select_dtypes(include=['object']).columns:
    X[col] = X[col].astype(str)
    X[col] = le.fit_transform(X[col])

# Scale continuous inputs to uniform [0, 1] range
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# 80/20 Stratified Split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)

# Train the Random Forest Model
rf = RandomForestClassifier(random_state=42, n_estimators=100,
max_depth=15)
rf.fit(X_train, y_train)

# Evaluate on 5,000 blind test samples
y_pred = rf.predict(X_test)
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
```

Chapter 5: Performance Results & BI Analytics

5.1 Predictive Metrics and Confusion Matrix

Following the execution of the Random Forest model on the isolated 5,000-sample test set, the classifier achieved an exceptional **Accuracy of 99.76%**. The precise classification mapping across the three engine safety tiers is detailed in the confusion matrix below (Table 5.1).

True Class \ Predicted	Safe	Risky	Critical Failure
Safe	**4,303**	1	0
Risky	3	**369**	6
Critical Failure	1	1	**316**

Table 5.1: Multi-Class Confusion Matrix evaluated on the 5,000-sample test set.

The Precision, Recall, and F1-Score metrics universally registered between 0.98 and 1.00. This near-perfect predictive capability is strongly justified: the algorithm did not overfit to noise; rather, it successfully deduced the **deterministic physical boundaries** embedded in the dataset. By identifying the exact thresholds where boost pressure mathematically overpowers the structural integrity of OEM internals under specific fuel types, the model accurately isolated critical failures.

5.2 DAX Integration and Business Intelligence

Within the Power BI dashboard, custom Data Analysis Expressions (DAX) were authored to calculate true context-aware averages, overriding the software's default aggregation behavior which historically inflated financial metrics by executing sum totals across vehicle rows.

```
Average_Build_Cost =
CALCULATE(
    AVERAGE('The_Flawless_Tuning_ERP_V20_Fixed'[Total_Build_Cost_USD]),
    FILTER(
        'The_Flawless_Tuning_ERP_V20_Fixed',
        'The_Flawless_Tuning_ERP_V20_Fixed'[Total_Build_Cost_USD] > 0
    )
)
```

The BI analysis successfully achieved the project's strategic objectives. It visualizes that transitioning from basic bolt-on modifications to fully forged engine internals significantly escalates capital expenditure (up to the $50,433 maximum), highlighting a clear mid-range "sweet spot" for horsepower ROI. Furthermore, interactive cross-filtering confirms the thermodynamic superiority of E85 Ethanol, proving it acts as a chemical intercooler that safely pushes power thresholds without requiring immediate, high-cost engine block reinforcements.

References

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